

New Chiral Fluoroanthryl Derivatives: Resolution of the Enantiomers by Chromatography on Cellulose Esters and Their Evaluation as Chiral Solvating Agents in NMR Spectroscopy

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ABSTRACT 2,2,2-Trifluoro-1-(9-anthryl)ethanol (TFAE) has now been widely used as a powerful chiral solvating agent for NMR spectroscopy. In connection with the development of a new general synthesis of halogenoalkylalkanols, starting from the corresponding ketone or aldehyde, we synthesized some halogenoalkyl-1-(9-anthryl)methanol derivatives liable to work as chiral solvating agents. The racemic anthryl derivatives were preparatively resolved into their corresponding enantiomers by chromatography on triacetyl cellulose (CTA I) or on *meta*-methylbenzoyl cellulose beads as chiral stationary phases. Their effectiveness as chiral solvating agents was measured as the magnitude of the splitting induced in the ^1H -NMR spectra of 1-phenylethylamine and of (1-phenylethyl)methyl ether in comparison with splitting caused by TFAE. While TFAE induced the largest splitting for 1-phenylethylamine, 2,2,3,3,3-pentafluoro-1-(9-anthryl)propanol 2 was more effective in the case of (1-phenylethyl)methyl ether, pointing out that depending on the substrate, other derivatives of the TFAE type can be very useful as chiral solvating agents.

KEY WORDS: cellulose triacetate, tribenzoylcellulose, methylbenzoylcellulose, chiral phase

INTRODUCTION

During the last decade, a growing number of publications in the field of organic chemistry have been dealing with the enantioselective synthesis, i.e., the preparation of chiral compounds in a nonracemic form. In connection with this development the determination of the enantiomeric excess has become a very important analytical task providing a quantitative measurement of the success of any enantioselective transformation.

Different techniques are available (polarimetry, isotope dilution, NMR spectroscopy, gas or liquid chromatography on chiral stationary phase, etc.) to determine the enantiomeric composition of a mixture and have recently been reviewed by Schurig.¹ Among those methods, NMR spectroscopy in the presence of chiral additives is commonly used and has been successfully applied to a large variety of substrates.^{2,3} These additives are either paramagnetic chiral lanthanide shift reagents or diamagnetic chiral solvating agents. Since the lanthanide shift reagents can lead to considerable line broadening, especially at the higher field strengths used nowadays, the older^{4,5} chiral solvating agents have enjoyed a strong renaissance. A very efficient chiral solvating agent developed by Pirkle is 2,2,2-trifluoro-1-(9-anthryl)ethanol 1 (TFAE).²

First resolved by Pirkle and Finn on a chiral station-

ary phase derived from dinitrobenzoyl phenylglycine ionically bonded to silica,⁶ the enantiomers of TFAE have the ability to be separated chromatographically on a large variety of totally different chiral stationary phases, such as cellulose triacetate,⁷ cellulose carbamate derivatives,^{8,9} poly(triphenylmethyl)methacrylate,^{10,11} amylose derivatives,¹² chiral polysiloxanes,¹³ quinine based phases,¹⁴ and various stationary phases based on amino acid derivatives.¹⁵⁻¹⁹

Since we found a general method for the preparation of arylhalogenoalkyl alkanols,^{20,21} we prepared a series of analogues of TFAE with different halogenoalkyl groups. The separation of the enantiomers of these alcohols by chromatography on various chiral stationary phases (CSPs) based on cellulose derivatives prompted us to evaluate the effectiveness of the enantiomerically pure alcohols as chiral solving agents compared with TFAE.

MATERIALS AND METHODS

Syntheses

The preparations of the racemic alcohols 1-4 have been carried out in analogy to the synthesis of 2,2-dichloro-3,3,3-trifluoropropanol reported earlier.^{20,21}

Received for publication January 3, 1991; accepted January 24, 1991.
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Anthracene-9-carbaldehyde is available from Fluka AG (Switzerland), and all perhaloalkanes used can be purchased from Fluorchem Ltd. (England).

In a typical experiment, a mixture of 0.10 mol of anthracene-9-carbaldehyde and 0.11 mol of perhaloalkane and 0.125 g-atom of freshly activated zinc²² in 180 ml of dry dimethylformamide (DMF) was vigorously stirred under argon at room temperature in a glass autoclave. After most of the zinc was consumed (usually between 1 to 24 h) the mixture was poured on a stirred slurry of ice, 10% aq. HCl and ether. Then the organic layer was separated, washed, evaporated, and the desired product purified either by column chromatography or by recrystallization.

The isolated yields of racemic 1–4 are (m.p. in parentheses) 1: 13% (136–138°C); 2: 50% (92–94°C); 3: 65% (116–117°C); 4: 51% (89–90°C).

Chromatographic Conditions

The chiral stationary phase cellulose triacetate (CTA I) has been prepared as previously reported.²³ The analytical CTA I glass column (column A, 1.25 × 25 cm, flow rate 0.5 ml/min) and the preparative column (column B, 5 × 100 cm) used in this work have already been described.²⁴ The mobile phase is always the mixture ethanol–water 95:5 when not otherwise specified.

The preparation of the benzoyl^{25,26} and methylbenzoyl cellulose^{26,27} stationary phases used in this work has also been reported previously. The analytical HPLC columns (0.46 × 25 cm) have been obtained by slurry packing of a suspension of the respective benzoylcellulose derivatives in hexane–2-propanol 90:10 at a pressure of 100 bar. The chromatographies have been performed using the same mixture as a mobile phase at a constant flow rate of 1 ml/min and at room temperature. The preparative column (column C, 5 × 75 cm) of *meta*-methylbenzoyl cellulose (MMBC) has been prepared analogously by slurry packing under low pressure conditions (10 bar) using hexane–2-propanol 90:10 as a mobile phase.

Preparative Resolutions

The racemic compounds 2–4 have been preparatively resolved in one run by injection of a quantity varying between 0.2 and 2 g on the respective preparative glass columns (column B, 320 ml/h, mobile phase: ethanol–water–cyclohexane 94:5:1 and column C, 720 ml/h mobile phase: hexane–2-propanol 90:10). The isolation and the characterization of the enantiomers (optical rotation in degrees and at 25°C) are given below.

Resolution of racemate 2

For the resolution of 2, 0.8 g of 2 in 10 ml of ethanol has been injected on column B containing CTA I (separation factor α , 1.99 and resolution factor R_s , 1.37); first eluted enantiomer, 1450–1900 ml: 370 mg (ee 100%), optical rotation (c 1.01 chloroform) $[\alpha]_D + 30.8$, $[\alpha]_{578} + 32.2$, $[\alpha]_{546} + 36.4$, $[\alpha]_{436} + 50.6$; last eluted enantiomer, 2050–2550 ml: 320 mg (ee > 99%), optical

rotation (c 1.05 chloroform) $[\alpha]_D - 30.6$, $[\alpha]_{578} - 32.1$, $[\alpha]_{546} - 36.4$, $[\alpha]_{436} - 50.3$.

Resolution of racemate 3

For the resolution of 3, 2 g of 3 in 30 ml of ethanol has been injected on column B containing CTA I (separation factor α , 3.26 and resolution factor R_s , 1.57); first eluted enantiomer 1770–2350 ml: 0.73 g (ee > 99%), optical rotation (c 0.53 chloroform) $[\alpha]_D + 88.3$, $[\alpha]_{578} + 93.2$, $[\alpha]_{546} + 108.9$, $[\alpha]_{435} + 222.2$; last eluted enantiomer, 4500–7300 ml: 0.72 g (ee 100%), optical rotation (c 0.52 chloroform) $[\alpha]_D - 92.7$, $[\alpha]_{578} - 97.5$, $[\alpha]_{546} - 113.3$, $[\alpha]_{436} - 228.3$.

Resolution of racemate 4

For the resolution of 4, 210 mg of 4 in 5 ml of hexane/2-propanol (mixture 9:1) has been injected on column C containing MMBC (separation factor α , 1.46 and resolution factor R_s , 1.78); first eluted enantiomer, 3000–4150 ml: 100 mg (ee 100%), optical rotation (c 1.59 chloroform) $[\alpha]_D - 69.8$, $[\alpha]_{578} - 73.6$, $[\alpha]_{546} - 85.8$, $[\alpha]_{436} - 171.6$; last eluted enantiomer, 4550–5500 ml: 95 mg (ee > 99%), optical rotation (c 1.42 chloroform) $[\alpha]_D + 66.3$, $[\alpha]_{578} + 69.9$, $[\alpha]_{546} + 80.9$, $[\alpha]_{436} + 162.7$.

NMR Experiments

The NMR spectra were measured with a Bruker AM 300 NMR spectrometer at room temperature using 0.7 ml of a 0.2% w/v solution of 1-phenylethylamine or (1-phenylethyl)methyl ether in CDCl₃ containing 100 mg of chiral solvating agent.

RESULTS AND DISCUSSION

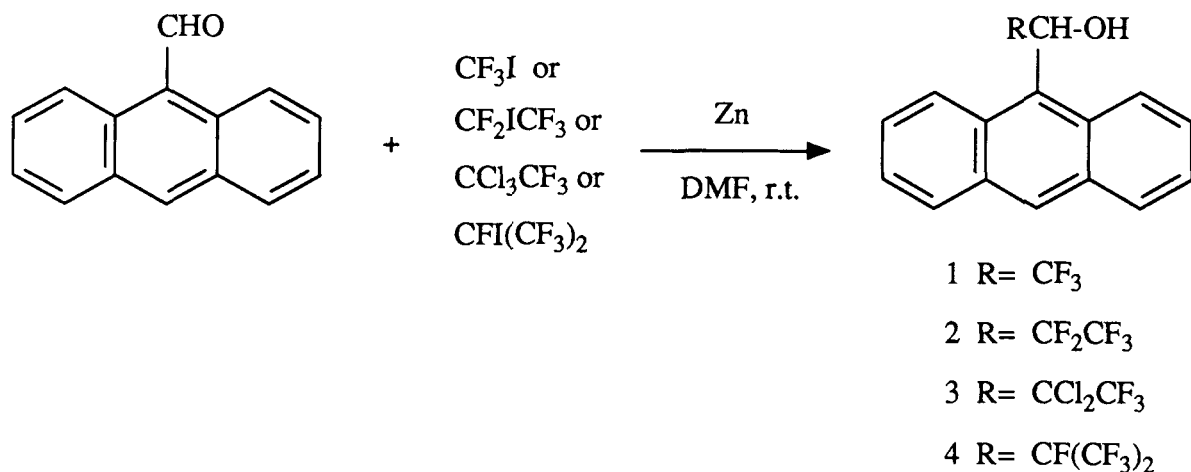
Syntheses

In analogy to earlier reports,^{20,21} all the racemic alcohols shown in Scheme 1 can be easily prepared starting from anthracene-9-carbaldehyde by reaction with the corresponding halogenoalkane in presence of zinc in dimethylformamide. The racemates are obtained in one step and with moderate yields.

Chromatographic Separations of the Enantiomers

The analytical separation of the enantiomers of compound 1 has been reported on different chiral stationary phases including cellulose derivatives, and they were separated on a preparative scale by Pirkle and Finn on the silica bonded dinitrobenzoyl phenylglycine phase that they introduced some years ago.⁶ They also reported the analytical separation of the enantiomers of 2 on the same dinitrobenzoyl phenylglycine CSP.⁶

In connection with our work on the development of chiral stationary phases for preparative separations we were interested in the separation of the enantiomers of the racemic fluoroanthryl derivatives 2–4, prepared according to the process described above, since these compounds were potentially chiral solvating agents. As chiral stationary phases we used four different cellulose esters; cellulose triacetate (CTA I) that we already investigated before,²³ and three benzoyl cellulose derivatives that we recently introduced, tribenzoyl cellu-



Scheme 1.

lose beads (TBC),^{25,26} *meta*-methylbenzoyl cellulose beads (MMBC) and *para*-methylbenzoyl cellulose beads (PMBC).^{26,27}

The analytical results are summarized in Table 1. On CTA I only 4 [R = $\text{CF}(\text{CF}_3)_2$] is rapidly eluting and shows no resolution. In contrast, 3 [R = CCl_2CF_3] exhibits the best resolution ($\alpha = 4.77$) of all the separations and is not at all or only poorly resolved on the benzoylcellulose derivatives. Surprisingly no separations is observed for any racemate on the nonsubstituted benzoyl cellulose TBC. Nevertheless good separations are obtained for 1 and 4 on the methylbenzoyl cellulose derivatives MMBC and PMBC. Compound 2 is resolved on CTA I, MMBC, and PMBC with a separation factor α ranging between 2.74 (CTA I) and 1.29

(MMBC). The inability of benzoyl cellulose for resolving the racemates 1–4 is contrasting with the good performances obtained with the methylbenzoyl cellulose derivatives, which differ only by the presence of a methyl group in the *meta* or *para* position of the aromatic ring. Furthermore, an inversion of the elution order is observed for compound 1 between *meta*-methyl and *para*-methyl benzoyl cellulose, indicating that even very similar derivatives of cellulose (differing only by the position of the methyl group on the aromatic ring) can exhibit completely different properties in terms of chiral recognition. This phenomenon has been observed for other racemates²⁷ and points out again the high potential of cellulose for the chromatographic separation of enantiomers. Indeed, very slight

TABLE 1. Chromatographic resolutions of 1–4 on cellulose derivatives

Racemate R	Stationary phase											
	CTA I ^{a,b}			TBC ^{c,d}			MMBC ^{d,e}			PMBC ^{d,f}		
	k'_2 ^g	α ^g	<i>h</i>	k'_2 ^g	α ^g	<i>h</i>	k'_2 ^g	α ^g	<i>h</i>	k'_2 ^g	α ^g	<i>h</i>
1 CF_3	2.08	2.47	(R)-(–)	5.92	1.00	<i>i</i>	38.14	1.84	(R)-(–)	22.80	1.18	(S)-(+)
2 CF_2CF_3	0.52	2.74	(–)	2.29	1.00	<i>i</i>	6.34	1.29	(+)	12.23	1.42	(+)
3 CCl_2CF_3	1.67	4.77	(–)	2.95	1.00	<i>i</i>	5.74	1.25	(+)	11.11	1.09	(+)
4 $\text{CF}(\text{CF}_3)_2$	0.14	1.0	<i>i</i>	1.57	1.00	<i>i</i>	3.08	2.05	(+)	4.00	1.48	(+)

^aCellulose triacetate.

^bMobile phase, ethanol–water, 95:5.

^cTribenzoylcellulose beads.

^dMobile phase, hexane–2-propanol, 90:10.

^e*Meta*-methylbenzoyl cellulose beads.

^f*Para*-methylbenzoyl cellulose beads.

^g k'_2 , capacity factor of the last eluted enantiomer; α , separation factor.

^hAbsolute configuration and/or sign of the optical rotation (λ , 436 nm) of the more retained enantiomer.

ⁱNot detectable.

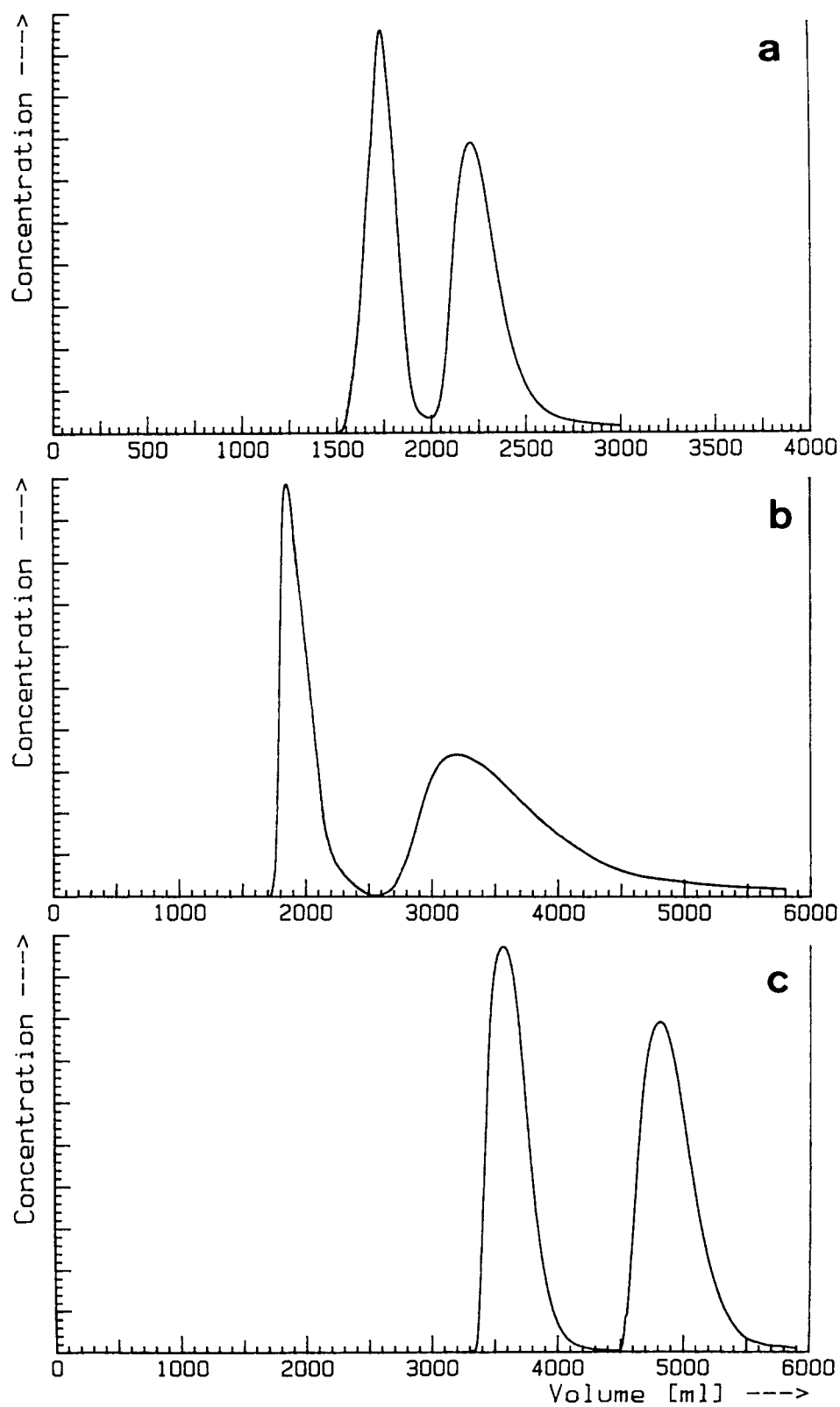
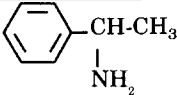
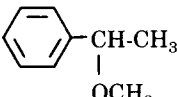


Fig. 1. Preparative chromatographic resolutions of (a) 0.8 g of racemate 2 on cellulose triacetate (column B), (b) 2 g of racemate 3 on cellulose triacetate (column B), and (c) 210 mg of racemate 4 on MMBC (column B).

TABLE 2. Splittings in ppm induced by 100 mg of chiral solvating agent 1–4 in the aliphatic signals in the ^1H -NMR spectra (CDCl_3) of 1-phenylethylamine and (1-phenylethyl) methyl ether

			Chiral solvating agent			
			1 CF_3	2 CF_2CF_3	3 CCl_2CF_3	4 $\text{CF}(\text{CF}_3)_2$
	CH		0.1002	0.0770	0.0464	0.0933
	CH_3		0.0154	0.0038	0.0023	0.0151
	CH		0.0205	0.0217	0.0143	0.0148
	OCH_3		0.0333	0.0463	0.0357	0.0325
	C-CH_3		0.0064	0.0112	0.0090	0.0106

modifications of the derivatizing group can completely alter the resolution power, giving from the same very inexpensive starting material chiral stationary phases exhibiting complementary chiral recognitions.

In order to evaluate their effectiveness as chiral solvating agents, compounds 2 and 3 have been preparatively resolved into the respective enantiomers on CTA I, while the enantiomers of 4 have been separated on MMBC. The separation of the enantiomers of 1 on CTA I on a pilot scale (up to 200 g) has been performed as we already previously reported.²⁸ The chromatogram of the preparative separations of 2–4 is shown in Figure 1.

NMR Experiments

The effectiveness of the chiral solvating agents was measured as the splitting induced in the aliphatic signals in the ^1H -NMR spectra of 1-phenylethylamine or (1-phenylethyl)methyl ether in comparison with the splittings caused by TFAE. The results summarized in Table 2 show that 4 induces splittings of a magnitude similar to that caused by TFAE for phenylethylamine, whereas 2 and 3 are poorer chiral solvating agents for this substrate. Nevertheless, in the case of (1-phenylethyl)methyl ether, compound 2 ($\text{R} = \text{CF}_2\text{CF}_3$) exhibits the largest splittings, clearly indicating that the effectiveness as the chiral solvating agent is substrate dependent. The origin of this substrate dependence probably arises from the slightly different acidity of 1–4 as already discussed by Pirkle and Hoekstra for 2,2,3,3,4,4,4-heptafluoro-1-phenylbutanol as compared with 2,2,2-trifluoro-1-phenylethanol.²⁹ The investigation of a larger number of substrates could possibly lead to a rationalization of this dependence.

CONCLUSION

This study showed that the new chiral solvating agents 2–4 can easily be prepared as racemates and separated into their enantiomers by chromatography on chiral stationary phases derived from cellulose. However, the selectivity of these chiral phases is strongly varying, depending on the halogenoalkyl group attached to the anthryl hydroxymethyl radical

common to the series of substrates 1–4. The effectiveness of 4 as chiral solvating agent is comparable to that of TFAE for phenylethylamine, but compound 2 ($\text{R} = \text{CF}_2\text{CF}_3$) is a stronger chiral solvating agent for (1-phenylethyl)methyl ether. These results point out the potential of the fluoroanthryl alcohol derivatives as chiral solvating agents and, owing to the observed substrate dependence, that their application field could probably be enlarged by using more systematically further analogues of TFAE.

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